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In [ ]:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
```

In [ ]:

```
file = "wireless_signal_kmeans_dataset.csv"
```

In [ ]:

```
data = pd.read_csv(file)
```

In [ ]:

```
import pandas as pd

#Ensure'file'and'data'areinitialized to prevent NameError
try:
    data
except NameError:
    file = "wireless_signal_kmeans_dataset.csv.csv"
    data = pd.read_csv(file)

if len(data.columns) == 1:
    print("CSV format issue detected. Repairing...")
    data = pd.read_csv(
        file,
        sep=",",
        engine="python"
    )

if len(data.columns) == 1:
    data = data.iloc[:,0].str.split(
        ",",
        expand=True
    )
    data.columns = [
        "RSSI",
        "RSRP",
        "RSRQ",
        "SNR",
        "Latency",
        "Packet_Loss
    ]    "

data.columns = data.columns.str.strip()
```

```
CSV format issue detected. Repairing...
```

In [ ]:

```
print("\nDataset Columns:")
print(data.columns)


print("\nDataset Preview:")
print(data.head())
```

```
Dataset Columns:
Index(['RSSI', 'RSRP', 'RSRQ', 'SNR', 'Latency', 'Packet_Loss'], dtype='obje
ct')

Dataset Preview:
  RSSI RSRP RSRQ SNR Latency Packet_Loss
0  -45  -75  1  -5    30      10             1
-48 -78 2 -50  -6    28      12             1
-80 3 -52 -82  -7    27      15             2
4 -55 -85      -8    25      18             2
               -9 23         20             3
```

In [ ]:

```
data = data.dropna()

for col in data.columns:

    data[col] = pd.to_numeric(
        data[col],
        errors="coerce"
    )

data.dropna(inplace=True)

print("\nClean Dataset:")
print(data.head())
```

```
Clean Dataset:
   RSSI    RSRP RSRQ SNR Latency Packet_Loss
0   -45   -75     -5   30      10           1
1   -48   -78     -6   28      12           1
2   -50   -80     -7   27      15           2
3   -52   -82     -8   25      18           2
4   -55   -85     -9   23      20           3
```

In [ ]:

```
X = data[
    [
        "RSSI",
        "RSRP",
        "RSRQ",
        "SNR",
        "Latency",
        "Packet_Loss
    ]    "
]
```

In [ ]:

```
scaler = StandardScaler()

X_scaled = scaler.fit_transform(
    X
)
```

In [ ]:

```
kmeans = KMeans(

    n_clusters=3,

    random_state=42,

    n_init=10
```

```
)

data["Cluster"] = kmeans.fit_predict(
    X_scaled
)
```

In [ ]:

```
print("\nCluster Count:")

print(
    data["Cluster"].value_counts()
)
```

```
Cluster Count:
         Cluster
0              11
2              10
1               8
Name: count, dtype: int64
```

In [ ]:

```
cluster_strength = data.groupby(
    "Cluster"
)["RSSI"].mean()

print("\nAverage RSSI:")

print(cluster_strength)
```

```
Average RSSI:
Cluster
0    -64.09090
1    9
2    -76.00000
    0
Name: RSSI, dtype: float64
    -50.50000
    0
```

In [ ]:

```
sorted_cluster = cluster_strength.sort_values(
    ascending=False
)

quality = {
    sorted_cluster.index[0]: "Good",
    sorted_cluster.index[1]: "Medium",
    sorted_cluster.index[2]: "Poor"
}

data["Signal_Quality"] = data["Cluster"].map(
    quality
)
```

In [ ]:

```
print("\nSignal Quality Groups:")

print(
    data[
        [
            "RSSI",
            "Cluster",
            "Signal_Quality"
        ]
    ]
```

```
)
```

```
Signal Quality Groups:
    RSSI Cluster Signal_Quality
0  1  -45        2         Good
2  3  -48        2         Good
4  5  -50        2         Good
6  7  -52        2         Good
8  9  -55        2         Good
10    -58        0       Medium
11    -60        0       Medium
12    -62        0       Medium
13    -65        0       Medium
14    -68        0       Medium
15    -70        0       Medium
16    -72        1         Poor
17    -75        1         Poor
18    -78        1         Poor
19    -80        1         Poor
20    -47        2         Good
21    -54        2         Good
22    -59        0       Medium
23    -64        0       Medium
24    -69        0       Medium
25    -74        1         Poor
26    -79        1         Poor
27    -46        2         Good
28    -51        2         Good
      -57        2         Good
      -63        0       Medium
      -67        0       Medium
      -73        1         Poor
      -77        1         Poor
```

In [ ]:

```
pca = PCA(
    n_components=2
)

X_pca = pca.fit_transform(
    X_scaled
)
```

In [ ]:

```
plt.figure(figsize=(8, 6))

sns.scatterplot(
    x=X_pca[:, 0],
    y=X_pca[:,                    1],
    hue=data["Signal_Quality"],
    palette="Set1",
    s=100
)

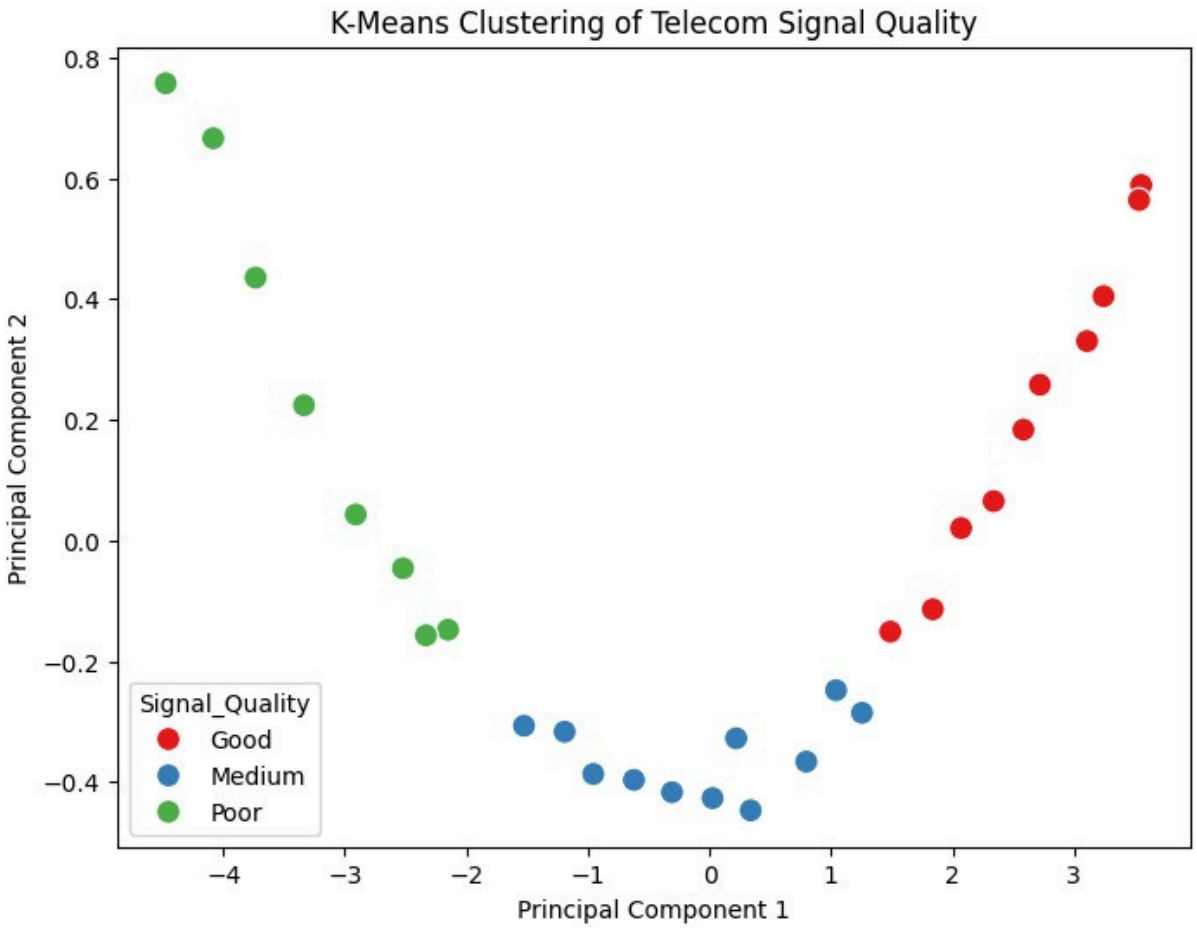
plt.title(
    "K-Means Clustering of Telecom Signal Quality"
)

plt.xlabel(
    "Principal Component 1"
)

plt.ylabel(
    "Principal Component 2"
```

```
)

plt.show()
```



In [ ]:

```
centers = scaler.inverse_transform(
    kmeans.cluster_centers_
)

print("\nCluster Centers:")

print(
    pd.DataFrame(
        centers,
        columns=X.columns
    )
)
```

Cluster Centers:

	RSSI	RSRP	RSRQ	SNR	Latency	Packet_Loss
0	-64.090909	-94.090909	-12.363636	15.000	42.818182	7.636364
1	-76.000000	-106.000000	-17.125000	5.625	98.750000	19.375000
2	-50.500000	-80.500000	-7.000000	26.400	15.400000	2.000000

In [ ]:

```
new_signal = [[
    -55,    # RSSI
    -85,    # RSRP
    -9,     # RSRQ
    23,     # SNR
    20, 3   # Latency
            # Packet Loss
]]
```

In [ ]:

```
new_scaled = scaler.transform(
    new_signal
)

cluster = kmeans.predict(
    new_scaled
)

print("\nNew Signal Cluster:")

print(cluster[0])

print(
    "Signal Quality:",
    quality[cluster[0]]
)
```

```
New Signal Cluster:
2
Signal Quality: Good /usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: Us erWarning: X does not have valid feature names, but StandardScaler was fitte d with feature names

  warnings.warn(
```

In [ ]: